Luke Rodel

Mechanical design · Automation · Manufacturing

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Hands-on engineering that connects mechanical design to working systems.

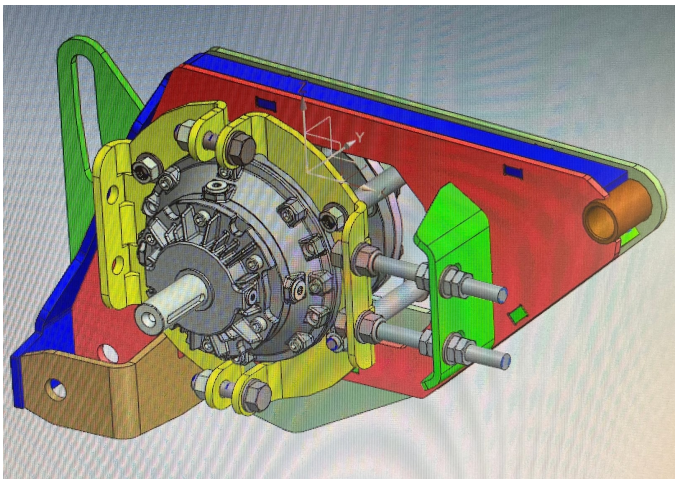
University of Alabama · B.S. Mechanical Engineering candidate · Expected May 2027

01 / SPUDNIK EQUIPMENT · SUMMER 2024

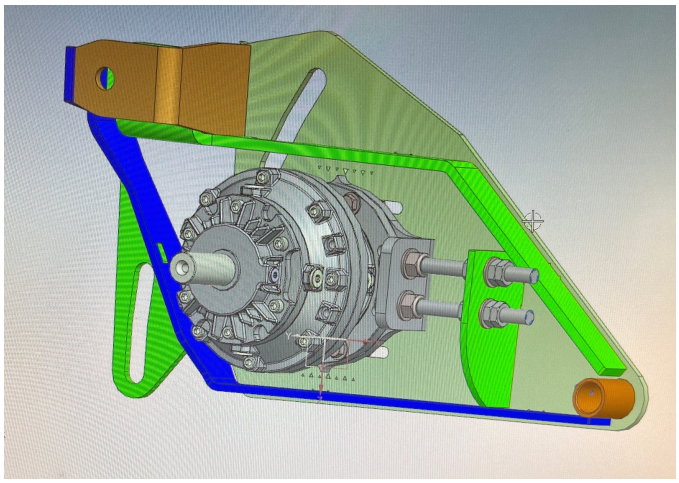
A simpler mount.
A more reliable drivetrain.

I redesigned a sugar beet harvester's chain-drive layout and mounting arrangement to improve tensioning, assembly, and service access.

> **10 fewer parts**
> in the redesigned assembly



BEFORE Original mounting assembly



AFTER My simplified design

The decisions

I investigated chain skipping and whip, iterated layouts in NX, raised the chain's low point, and incorporated tensioning arms. Simplifying the mount improved assembly access.

The validation

After implementation, the system ran at speed without the previously observed skipping. The website includes the original running clip and more detail about my role.

NX CAD · Design for assembly · Chain tensioning · Hardware validation

Currently building: robotic-arm direct ink writing capstone for non-planar polymer lattices. In progress.

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02 / MERCEDES-BENZ · FALL 2024



Inspection without the extra steps.

I tested scanning angles and designed an angled printer stand to reduce glare. Alongside the robotics lead, I helped integrate automatic grading as labels printed.

Approx. 10 seconds

saved in the label workflow

Print > inspect > accept or reprint

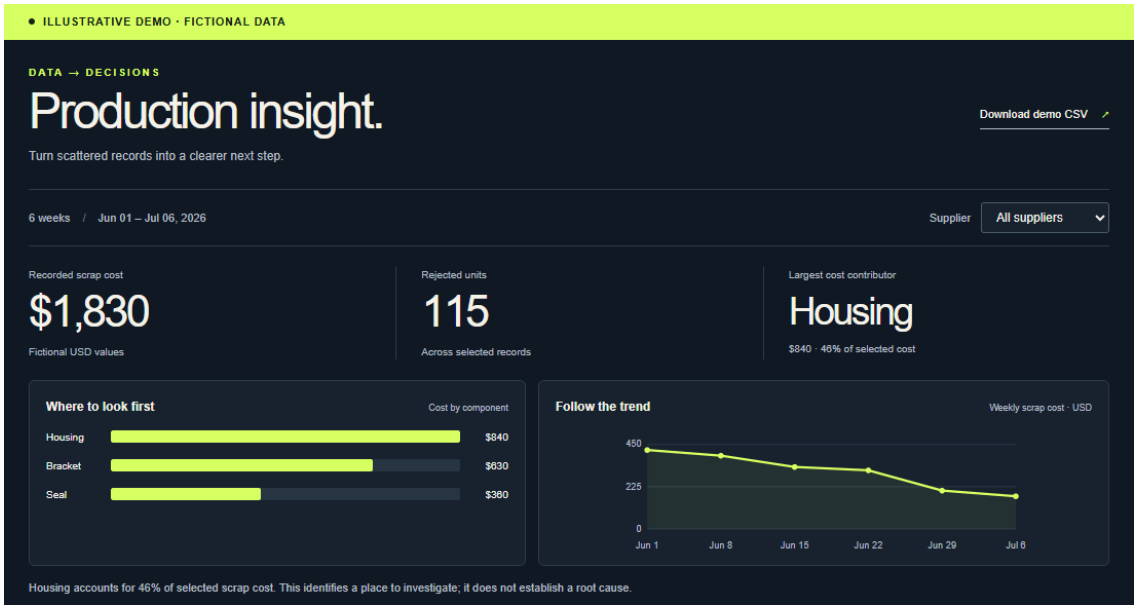
03 / MERCEDES-BENZ · SUMMER 2025

A clearer picture of production.

I built Power BI tools and mapped defect, repair, reorder, and chargeback workflows across departments.

Hours to minutes

for a separate monthly reporting workflow



Independent web recreation with fictional data. This is not the employer's Power BI report.

I used Power Query / M and DAX to prepare data and improve visibility. A related process investigation informed a proposal for better scrap and reclaim traceability; it is distinct from an implemented plant-wide system or a verified financial saving.